

Problem (or opportunity): LIs do sometimes get tricked

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Consider an infinite Decision Tree (that is, an Ω) such that, at every node, our agent $\mathbf{A}$ can choose between Cooperate and Defect. If Cooperate, it receives 0 utilons. If Defect, it receives $1/\mathbb{P}_n(D_n)$ utilons (where n is the current round). $\mathbb{P}$ is Ω 's logical inductor, that knows $\mathbf{A}$'s algorithm, and is fed all past observations (both the actions of $\mathbf{A}$ and Ω 's responses).¹ D_n is a sentence encoding " $\mathbf{A}$'s algorithm Defects in round n ".

It might seem like the highest possible limit average reward per round we could get is 1. Indeed, if $\mathbf{A}$ Defects with pseudorandom probability p/q (and does so by consulting complex enough pseudorandom coins, so that $\mathbb{P}$ can't predict them in advance), then in the limit $\mathbb{P}$ will catch on to that reality, and the average reward will be $\frac{p \cdot \frac{q}{p}}{q} = 1$. $\mathbf{A}$ can do fancier stuff, like choosing ever-unlikelier pseudorandom coins. But $\mathbb{P}$ will always end up assessing these probabilities as well as $\mathbf{A}$ does (that is, "noticing the pattern of what $\mathbf{A}$ is doing"), and so $\mathbf{A}$ won't get over 1 in expectation.

Suppose instead $\mathbf{A}$ is the following algorithm: "Defect iff $1/\mathbb{P}_n(D_n) > 2n$ ". $\mathbb{P}$ will eventually "be aware" that $D_n \leftrightarrow 1/\mathbb{P}_n(D_n) > 2n$. And nonetheless, $1/\mathbb{P}_n(D_n) > 2n$ will happen infinitely often, and so $\mathbf{A}$ will have 2 as a limit point of its average reward (which nonetheless might not converge). Indeed, suppose otherwise that, after some m , $1/\mathbb{P}_n(D_n) > 2n$ is never the case again. That is, $\mathbb{P}_n(D_n) \geq 1/2n$. Then, a trader always selling 1 share of D_n each round after m exploits $\mathbb{P}$.

I don't yet know if $\mathbb{P}_n(D_n)$ converges to 0, nor whether the average reward converges (we could have to wait too long for the new Defection infinitely often, so that some number < 2 is also a limit point). Of course, the average reward converging (or at least not dropping below 1 in the limit) would seem more pathological. Otherwise, it's not as straightforward to label this an improvement for $\mathbf{A}$.

This phenomenon is interesting, but probably what we want to do for our paper is just "assume we can prove a priori bounds on the reward a single node provides". For example, using $\min(1/\mathbb{P}_n(D_n), 100)$ in this case.

¹This argument also works with an "action-based" LI: That is, Ω doesn't know $\mathbf{A}$'s algorithm in advance, and $\mathbb{P}$ just learns about a syntactic variable a_n . But it is "even more pathological" that it works for the usual setting.